Git Command-Line Cheat Sheet

Practical reference — frequent commands first, then important commands for history, recovery, and troubleshooting.

1. Commands You'll Use All the Time

What it does	Command
Check status	git status
Add everything	git add .
Add a specific file	git add filename
Commit	git commit -m "Your commit message"
Push	git push
Push explicitly	git push origin main
Pull	git pull
Clone	git clone <repository-url>
Commit history	git log --oneline
Visual history	git log --oneline --graph --decorate --all
Remote repository	git remote -v

2. Branches

What it does	Command
List branches	git branch
All local + remote branches	git branch -a
Create branch	git branch feature-name
Switch branch	git switch feature-name
Create + switch	git switch -c feature-name
Delete local branch	git branch -d feature-name
Force-delete branch	git branch -D feature-name # CAUTION

3. Undoing Things — Important

What it does	Command
Edit latest commit message	git commit --amend -m "Correct message"
Undo latest commit, keep changes staged	git reset --soft HEAD~1
Undo latest commit, keep changes unstaged	git reset HEAD~1
Undo latest commit and delete changes	git reset --hard HEAD~1 # DANGEROUS
Safely reverse a commit	git revert <commit-hash>

4. Fixing Push / Pull Problems

What it does	Command
Check status	git status
Fetch remote changes	git fetch
Pull with rebase	git pull --rebase
Push after a history rewrite	git push --force-with-lease # safer than --force

What it does	Command
Avoid unless you understand the consequences	git push --force # DANGEROUS

5. Rebase

What it does	Command
Rebase current branch onto main	git rebase main
Interactive rebase	git rebase -i HEAD~3
Interactive actions	pick = keep reword = change message edit = modify commit squash = combine with previous fixup = combine, discard message drop = remove commit

6. Git Config

What it does	Command
List configuration	git config --list
Set username	git config --global user.name "Your Name"
Set email	git config --global user.email "you@example.com"
Check username	git config user.name
Check email	git config user.email
Default initial branch	git config --global init.defaultBranch main

7. Inspecting Changes

What it does	Command
Unstaged changes	git diff
Staged changes	git diff --staged
Show a commit	git show <commit-hash>

8. Stash

What it does	Command
Temporarily save unfinished work	git stash
List stashes	git stash list
Restore latest stash and remove it	git stash pop
Restore without removing	git stash apply
Delete a stash	git stash drop

9. Tags

What it does	Command
Create tag	git tag v1.0
List tags	git tag
Push tag	git push origin v1.0
Push all tags	git push --tags

10. Commands Where You Should STOP & Think

What it does	Command
Hard reset	git reset --hard
Force push	git push --force
Force-delete branch	git branch -D branch-name
Rewrite history	git rebase -i HEAD~N

11. The Git Mental Model

Working Directory
    |
    | git add
    v
Staging Area
    |
    | git commit
    v
Local Repository
    |
    | git push
    v
GitHub / Remote

GitHub --git fetch--> remote information
GitHub --git pull--> local branch

12. The 15 Commands to Memorize First

git status
git add .
git commit -m "message"
git push
git pull
git clone <url>
git log --oneline
git log --oneline --graph --decorate --all
git branch
git switch <branch>
git switch -c <new-branch>
git fetch
git diff
git show <commit>
git revert <commit-hash>

Golden Rule

Before using **reset --hard**, **rebase**, **push --force**, or **branch -D**: STOP → run **git status** → inspect **git log --oneline --graph --decorate --all** → make sure you understand what will change.